Between Inspiration and Imitation: Graded Fashion Infringement Assessment with Deep Metric Learning and a Calibrated Decision Layer

Abstract

Fashion design infringement sits at an ambiguous boundary between lawful inspiration and illegal copyright, making it poorly suited for binary visual similarity systems. This study formulates fashion infringement assessment as a graded three-class classification task over *original*, *similar*, and *knockoff* cases. The implemented system follows a two-stage pipeline: a deep metric-learning component provides fashion-aware visual representations, and a calibrated downstream decision layer maps pairwise embedding-derived features into interpretable infringement categories and DMCA-style actions. Using a curated dataset of fashion design cases, cosine-threshold baselines are compared with learned classifiers operating on nearest-original comparison features. The strongest current result is a pairwise multilayer perceptron decision layer, which achieves a test macro-F1 of 0.7368 in the uncalibrated comparison setting. After temperature scaling, the calibrated deployment-oriented version achieves 0.6926 test macro-F1 while improving validation expected calibration error from 0.1042 to 0.0857. These results indicate that graded infringement assessment is better handled by a calibrated non-linear decision layer than by fixed similarity thresholds alone, and that separating representation learning from decision calibration provides a more interpretable and practically useful framework for computer-assisted fashion copyright review.

1. Introduction

Fashion design infringement is difficult to assess because the domain sits between artistic expression, functional design, and rapid commercial imitation. Many fashion copyright cases do not involve exact duplication, but rather ambiguous overlap in motifs, prints, proportions, or detailing. This creates a poor fit for binary knockoff/original systems. Legal analysis of fashion copyright has also emphasized that many design elements, including silhouettes, trends, and broad stylistic conventions, are only weakly protected or treated as functional rather than expressive, making the boundary between lawful inspiration and infringement especially difficult to prove and prosecute accordingly (17). In the UK, the procedural reality of such lower-value and less complex intellectual property disputes is also reflected in the specialist role of the Intellectual Property Enterprise Court (14).

This study investigates whether fashion infringement can be modeled more effectively as a graded decision problem. Rather than asking only whether two designs are originals or knockoffs, the study considers three categories: *Original*, *Similar*, and *Knockoff*. This framing reflects the practical reality that many fashion cases occupy an intermediate region of stylistic similarity that is neither clearly lawful inspiration nor clearly illegal copying.

This study is a two-stage pipeline that separates visual representation learning from downstream decision-making. First, a deep triplet metric-learning stage based on a ResNet encoder is used to learn fashion-aware embeddings from image data, as seen in previous studies with metric-learning literature on similarity-based representation spaces (1; 2; 3; 16). Second, a calibrated decision layer takes pairwise features derived from a candidate design and its nearest *original* reference and converts them into both class predictions and confidence-aware policy actions. At the prediction level, the model assigns one of the three labels *original*, *similar*, or *knockoff*. At the policy level, those calibrated probabilities are translated into operational outcomes: high-confidence predicted knockoffs can be *auto_flagged*, borderline or uncertain cases can be sent to *review*, and low-confidence non-critical cases can be assigned *no_action*. This differs from prior retrieval-style systems by focusing not only on similarity estimation, but on interpretable and calibrated decision support for ambiguous cases in which the main question is not simply "what is most similar?", but "what action should be taken given the model's confidence?"

The main research questions are:

1. Can embedding-based visual similarity support reliable distinction between *original*, *similar*, and *knockoff* fashion cases?
2. How sensitive are graded infringement decisions to the choice of thresholding versus learned non-linear decision layers?
3. Can deep metric learning on fashion imagery provide a stronger representation space for downstream infringement assessment?

The strongest current evidence comes from the multilayer perceptron calibrated decision layer rather than from a completed end-to-end evaluation of metric-learning variants. The experiments show that decision calibration and pairwise feature design are critical to the task, and that they provide a strong baseline against which future representation-learning improvements can be measured.

Compared with the CW3 plan, the emphasis of the final report therefore shifts slightly. The interim plan was to explore a broad comparison of metric-learning variants and downstream effects, whereas the classification results are strongest for the calibrated multilayer perceptron decision layer built on the present embedding pipeline. The final report reflects this change directly. The study treats the deep metric-learning component as the representation-learning foundation of the project, while presenting the most complete empirical findings around graded classification, calibration, and DMCA-style triage.

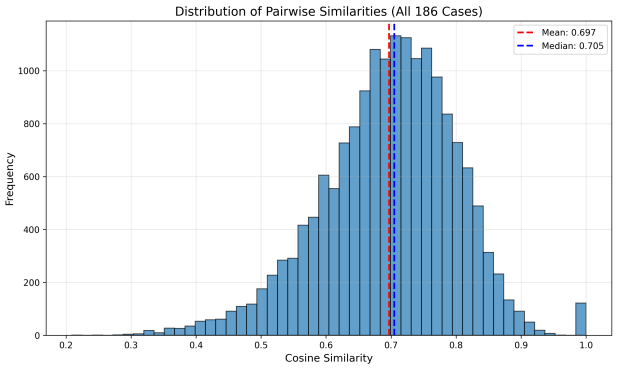

Figure 1. Similarity-score structure in the current dataset. The overlap between classes motivates treating infringement assessment as a graded decision problem rather than a single fixed-threshold retrieval task.

2. Task and Data

The primary task is graded fashion infringement classification. Each example is represented as an embedding associated with a fashion design case and compared against a nearest *original* reference design. The downstream classifier predicts one of three classes:

- **Original**: Visually and semantically distinct from the nearest protected design;
- **Similar**: Shares stylistic attributes but differs in expressive detail;
- **Knockoff**: Highly overlapping in expressive detail and likely copied.

The multilayer perceptron uses a dataset of 743 embedded comparisons with 512-dimensional feature vectors, split into 519 training examples, 110 validation examples, and 114 test examples using a fixed random seed. The class mapping is knockoff = 0, similar = 1, and original = 2.

In order to clean both the DeepFashion and gold cases datasets, cleaning scripts for constructing the evaluation split from curated and scraped fashion-design cases are in the code zip. The custom case embeddings are produced by encoding textual design descriptions with a frozen CLIP text encoder and then L2-normalizing the resulting 512-dimensional vectors. CLIP (Contrastive Language-Image Pre-training) is a multimodal model trained to align images and text in a shared embedding space. In this study, only the text encoder is used to convert design descriptions into fixed-length semantic representations (4). These embeddings are subsequently stabilized, split into train/validation/test partitions, and transformed into pairwise comparison features. Preprocessing for the decision-layer experiments includes nearest-original retrieval by cosine similarity and feature construction from the sample-to-reference difference. The resulting pairwise statistics features include cosine similarity, cosine distance, Euclidean distance, absolute-difference statistics, and related summary features.

For representation learning, the project uses DeepFashion as an image dataset of over 40,000 images to create a robust model for visually categorizing fashion items (5). DeepFashion does not contain infringement labels, but it provides textual item identity information that allows positive pairs or triplets to be formed from different views of the same garment and negatives to be drawn from different items. Before entering the encoder, each RGB image is resized to 256 pixels on the shorter side, center-cropped to 224×224, converted to a tensor, and normalized channel-wise using the standard ImageNet mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$. This keeps the input format aligned with the ImageNet pretraining regime of the ResNet backbone and allows the model to separate generic fashion representation learning from the smaller, more subjective infringement-classification layer.

The main evaluation metrics are classification accuracy and macro-averaged F1 score. Macro-F1 is especially important here because it evaluates class balance more fairly than accuracy alone and better reflects performance on the ambiguous middle class. For the calibrated policy analysis, the report additionally evaluates expected calibration error, confidence-threshold trade-offs, and knockoff auto-flag precision and recall.

3. Methodology

The study uses a two-stage methodology because similarity learning and infringement decision-making are related but distinct problems. The representation model should capture visual proximity between fashion items, while the final classifier must translate that proximity into stable and interpretable labels under legal and semantic ambiguity.

3.1. Representation Learning

The representation learning stage is implemented as a triplet-network pipeline with a shared-weight ResNet backbone. Three parallel paths with tied weights process the anchor, positive, and negative inputs with the same encoder parameters, so the network learns a single embedding function rather than separate class-specific predictors. The backbone can be either `resnet18` or `resnet50`, initialized from ImageNet-pretrained weights. The original classification layer is removed and replaced with a linear embedding head that maps the final pooled ResNet feature vector to a 128-dimensional representation. This gives a compact descriptor that is large enough to retain garment-level detail, while still being small enough for stable metric learning and nearest-neighbor comparison. The use of a residual convolutional backbone follows the standard ResNet formulation for deep visual feature extraction (9).

From an architectural perspective, the encoder is a standard convolutional residual network. The input image first passes through an initial large-kernel convolution, normalization, non-linearity, and spatial downsampling stage, which rapidly converts raw RGB pixels into low-level feature maps. These early convolutional filters typically respond to primitive visual structure such as edges, corners, color boundaries, and local texture changes. Subsequently, the network applies a sequence of residual stages that progressively increase channel depth while reducing spatial resolution. This coarse-to-fine hierarchy is useful for fashion imagery because it lets the model combine local evidence such as stitching, pleats, or print fragments with broader shape evidence such as neckline, sleeve geometry, skirt length, or overall silhouette.

The residual design is especially important in this study because it allows deeper convolutional feature extraction without making optimization unstable. Each residual block learns a transformation relative to an identity shortcut, so later layers can refine rather than completely overwrite earlier representations. Essentially, this helps the backbone preserve visually important garment signals across depth: lower layers retain local texture and contour information, intermediate layers assemble motifs and garment parts, and later layers encode more global configurations that are useful for item-level comparison. For infringement-style similarity assessment, this layered feature composition is preferable to using only shallow handcrafted similarity signals, because potential copying may appear in combinations of cut, trim placement, print layout, and proportion rather than in a single isolated detail.

Fine-tuning is deliberately restricted to the last residual block (`layer4`) and the new embedding head while earlier convolutional layers remain frozen. Essentially, the lower and middle stages of ResNet already encode broadly useful visual primitives such as edges, color transitions, repeated textures, and simple shape fragments, whereas the later layers are more task-specific and therefore better suited to adaptation toward fashion cues such as silhouette, print placement, seam structure, and surface detailing. This partial-freezing strategy also reduces the number of trainable parameters, which is important given the modest scale of the available fashion-identity training setup. Input images are resized to 256 pixels on the short side, center-cropped to 224×224, converted to tensors, and normalized with ImageNet channel statistics before being passed through the backbone. This design is consistent with standard deep metric-learning practice using Siamese or triplet formulations for similarity modeling (1; 2; 16).

The two supported Resnet backbones differ mainly in depth and representational capacity. `resnet18` is built from comparatively simple residual blocks and offers a lighter-weight encoder that is easier to train and less prone to overfitting in smaller-data regimes. `resnet50` is substantially deeper and uses bottleneck residual blocks, allowing richer hierarchical feature extraction at the cost of greater computational complexity. In the context of this study, this makes `resnet18` a pragmatic baseline and `resnet50` a higher-capacity alternative for capturing subtle garment-level distinctions. In both cases, the final global average pooling step collapses the spatial feature map into a single vector before the learned embedding head projects it into the metric space used by the triplet loss.

Triplets are constructed from textualized item identities in the image dataset. For each anchor image, a positive example is drawn from another image of the same item and a negative example is drawn from a different item. This means the supervision signal does not rely on infringement labels at this stage, only on the weaker assumption that two photographs of the same garment should lie nearby in the learned space, while different garments should be separated. After the forward pass, each branch output is L2-normalized, so comparison is based on angular similarity rather than raw feature magnitude. The model is trained with a cosine-distance triplet objective:

$$\mathcal{L} = \max(0, d(a, p) - d(a, n) + m),$$

$d(\cdot, \cdot)$ is cosine distance m is a margin hyperparameter. In this study, $m = 0.2$.

The loss penalizes the model whenever the negative is not at least a margin farther from the anchor than the positive. Intuitively, the encoder is therefore trained not to memorize item IDs directly, but to arrange the embedding space so that visually related garments cluster and unrelated garments are pushed apart with explicit geometric separation. Optimization uses Adam (15) with learning rate 10^{-4}, batch size 32, weight decay 10^{-4}, and 10 training epochs. These settings are deliberately conservative because the model is initialized from ImageNet-pretrained weights and only the later layers are fine-tuned. A small learning rate reduces the risk of destroying useful visual features too quickly, while weight decay provides regularization against overfitting. The batch size balances gradient stability against memory cost, and the limited number of epochs caters to the aim of adapting a strong visual backbone rather than fully training the model from scratch. Common practices undergo this

process when finetuning pretrained convolutional networks for downstream visual tasks (4).

3.2. Calibrated Decision Layer

The downstream classifier operates on pairwise features derived from the embedding of a candidate design and its nearest *original* reference in the training set. For each example, cosine similarity is used to identify the nearest reference original among the training items labeled *original*. This is to anchor every decision against the closest protected-style reference rather than against an arbitrary neighbor from any class. Features are then computed from the sample-reference pair. The main input is the element-wise absolute-difference vector $|x - r|$, which retains per-dimension disagreement information from the embedding space. This vector is augmented with scalar statistical summary features: cosine similarity, cosine distance, Euclidean distance, mean absolute difference, variance of absolute difference, maximum absolute difference, the 95th percentile of absolute differences, and the gap between the highest and second-highest reference similarities. The last quantity acts as a retrieval-stability feature. Essentially, a small gap indicates that the nearest-original match is ambiguous, while a large gap suggests a more decisive comparison.

Three decision strategies are evaluated:

1. A cosine-threshold baseline, which searches for two validation thresholds $t_1 < t_2$ to partition the examples into *Original*, *Similar*, and *Knockoff*

2. A logistic regression on scalar pairwise features

3. A multilayer perceptron (MLP) with two hidden layers of sizes `[256, 64]` applied to the concatenation of absolute-difference features and scalar pairwise features.

The MLP is trained as a Stage-2 classifier such that the classifier is a standardized feature pipeline followed by a feed-forward network trained with the default non-linear hidden activations of `sklearn`'s `MLPClassifier` and early stopping. Some dimensions come from the absolute-difference vector, while others are scalar distances or summary statistics on different scales. Thus, standardizing makes the learned decision boundary explicit and allows pairwise feature engineering, calibration, and error analysis to be studied independently from the embedding model.

For deployment-oriented analysis, the MLP probabilities are calibrated using temperature scaling on the validation set (6). The uncalibrated class-probability vector p is first converted to log-probabilities $\log p$. A scalar temperature T is then selected by grid search over validation values from 0.5 to 5.0 to minimize negative log-likelihood. The calibrated probabilities are computed as

$$\tilde{p}_k = \frac{\exp(\log p_k / T)}{\sum_j \exp(\log p_j / T)}.$$

When $T > 1$, the distribution is softened and overconfident predictions are reduced, while when $T < 1$, the distribution is sharpened. In the experiments, the ideal learned temperature is $T = 1.55$, which indicates that the raw MLP probabilities were somewhat overconfident on the validation split.

Calibration is kept separate from classification. It does not change the embedding features or retrain the MLP weights, but only rescales the output confidence scores to create operational interpretability. After temperature scaling, the predicted class is still the argmax of the calibrated distribution, but the confidence values become better aligned with empirical correctness. These calibrated probabilities are then mapped to policy actions through two thresholds. Cases predicted as *knockoff* with confidence at least τ_{auto} are assigned *auto_flag*, while remaining cases with confidence at least τ_{review} are assigned *review*. All other cases are assigned *no_action*. This turns the model into a triage mechanism to handle the ambiguity between knockoffs and similar designs, and is closely related to the broader idea of selective prediction or reject-option learning in uncertainty-aware classification (7; 13).

4. Experiments

The main research question this study explores is whether a graded infringement decision layer can be built from existing visual embeddings, and how much calibration improves its practical usefulness.

Experiment 1: Threshold Baseline. The first experiment evaluates whether simple similarity thresholds are sufficient. Two thresholds are selected on the validation set and then applied to the test set. This provides a transparent but deliberately limited baseline.

Experiment 2: Learned Linear Baseline. The second experiment uses logistic regression on scalar pairwise features. The purpose is to test whether a learned but linear boundary already improves on thresholding.

Experiment 3: Learned Non-Linear Decision Layer. The third experiment uses an Multilayer Perceptron over the concatenation of absolute-difference features and scalar pairwise statistics. This tests whether non-linear interactions between embedding-derived features are necessary to separate the ambiguous *similar* region from the *original* and *knockoff* classes.

Experiment 4: Calibration and Policy Analysis. The final experiment calibrates the Multilayer Perceptron outputs using temperature scaling and evaluates policy-oriented thresholds for *auto_flag*, *review*, and *no_action*. In practical fashion copyright cases, confidence-aware triage and false-positive control is essential, beyond raw classification accuracy.

All downstream classifier experiments use the same fixed train/validation/test split, the same nearest-original feature

construction, and the same macro-F1/accuracy metrics. As depicted in the below plots, evaluation metrics includes confidence intervals, confusion matrices, and exported error-case summaries to support analysis beyond headline scores.

The nearest-*original* comparison design is important to the problem formulation because the legal and practical question is not whether a candidate resembles any arbitrary example in the dataset, but whether it most closely approaches a protected-style reference that could plausibly ground an infringement concern. Restricting the reference set to training cases labeled *Original* therefore makes the comparison more interpretable and more aligned with the intended downstream use. It also creates a harder and more realistic decision problem than unconstrained nearest-neighbor matching, because the classifier must reason about the type of deviation from an original reference rather than merely about raw similarity to any nearby example.

The evaluation metrics are chosen to reflect both predictive performance and deployment suitability. Macro-F1 is treated as the primary discrimination metric because the intermediate *similar* class is the most ambiguous and would otherwise be underweighted by accuracy if the model performed well on the easier outer classes. Expected calibration error is included because the policy layer depends directly on confidence quality. A triage system that reports poorly calibrated probabilities would make threshold-based review decisions unreliable even if its top-1 class predictions were competitive. While Expected Calibration Error (ECE) is a useful summary measure in practice, calibration in multiclass settings is itself more subtle than a single scalar can capture (8), and broader work on calibrated probability estimation further motivates this emphasis on confidence quality (12). Together, macro-F1 and calibration provide a better account of practical system behavior than accuracy alone.

For reproducibility, the principal settings are as follows. The threshold baseline searches over validation-set cosine-distance quintiles to choose $t_1 < t_2$. Logistic regression is trained on standardized scalar pairwise features. The MLP uses two hidden layers of sizes `[256, 64]` with standardized inputs and up to 400 optimization iterations. Temperature scaling is fitted on the validation split and yields a learned temperature of 1.55. In the representation-learning code path, the triplet model uses an ImageNet-pretrained ResNet backbone, embedding dimension 128 by default, margin 0.2, learning rate 10^{-4}, batch size 32, 10 training epochs, and random seed 42. The reported MLP metrics in this report correspond to the current embedding pipeline based on frozen CLIP text encodings of case descriptions.

5. Results and Evaluation

5.1. Graded Infringement Classification

Table 1 summarizes the main uncalibrated comparison results. The cosine-threshold baseline achieves a test macro-F1 of 0.4809, indicating that rigid partitions in embedding-distance space are insufficient for the three-class problem. Logistic regression on pairwise features performs similarly poorly, with a macro-F1 of 0.4485. In contrast, the pairwise MLP achieves 0.7368 macro-F1 and 0.7368 accuracy, showing that a learned non-linear boundary over richer pairwise features can capture the ambiguity structure of the task much more effectively.

Model	Accuracy	Macro-F1
Cosine Thresholds	0.4825	0.4809
Logistic Regression (pairwise)	0.4474	0.4485
MLP (abs diff + pairwise)	**0.7368**	**0.7368**

Table 1. Stage-2 comparison on the held-out test set.

The threshold baseline assumes that infringement categories can be arranged as simple consecutive bands along a single cosine-distance axis. However, the lower macro-F1 result suggests that this assumption is too primitive as many cases that are close to an original are not necessarily knockoffs, and some knockoff-like cases are distinguishable only when the patttern of embedding differences is taken into account rather than their overall magnitude alone. Logistic regression improves flexibility slightly by combining scalar pairwise features, but it still imposes a linear boundary and therefore cannot model the interaction-heavy decision regions that arise when the *similar* class overlaps with both neighboring classes. The strong MLP macro-F1 therefore supports the claim that the task is not only about ranking similarity, but about learning a more structured decision function over multiple comparison signals.

The confusion matrices further support this interpretation. The threshold model produces substantial cross-class confusion, especially around the middle class. The MLP reduces this substantially, with the saved test confusion matrix

$$\begin{bmatrix} 29 & 6 & 5 \\ 4 & 27 & 6 \\ 3 & 6 & 28 \end{bmatrix},$$

using the label order `[knockoff, similar, original]`. Rows correspond to true labels and columns correspond to predicted labels. This indicates significantly better separation across all three regions of the decision space.

5.2. Calibration and DMCA Policy Analysis

After temperature scaling, the MLP achieves 0.6930 test accuracy and 0.6926 test macro-F1. Validation expected calibration error improves from 0.1042 before calibration to 0.0857 after calibration, indicating that the predicted probabilities better reflect empirical correctness.

The threshold baseline obtains a 95% confidence interval of [0.3873, 0.5650] for test macro-F1, while the calibrated MLP achieves [0.6034, 0.7710]. Although calibration slightly reduces raw classification performance relative to the best uncalibrated MLP model, it yields probabilities that are more useful for downstream confidence-aware policy

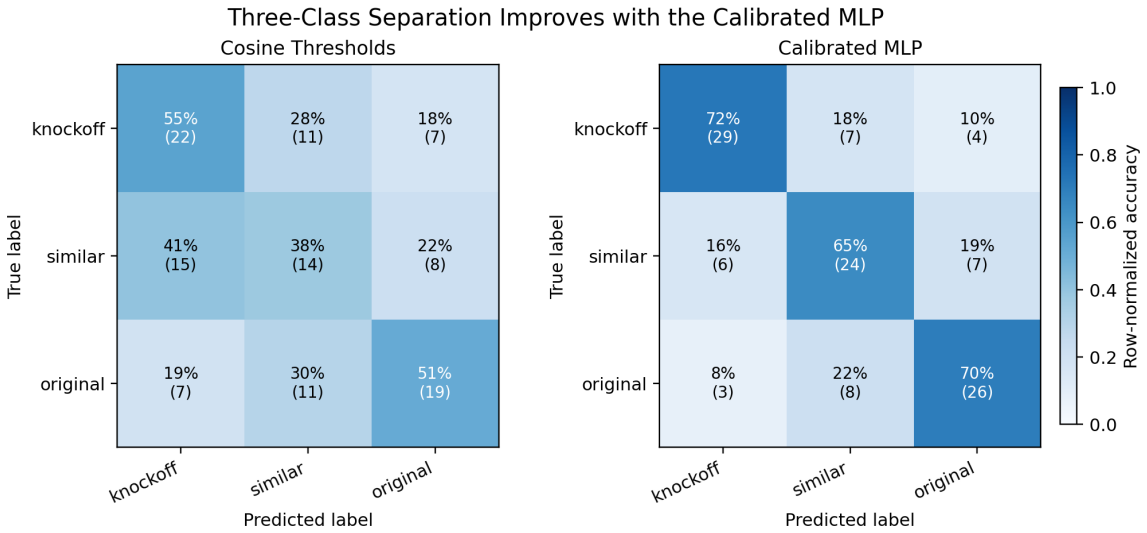

Figure 2. Row-normalized confusion matrices for the threshold baseline and the calibrated MLP on the test set. The calibrated MLP improves separation across all three classes, especially by reducing confusion involving the ambiguous *similar* category.

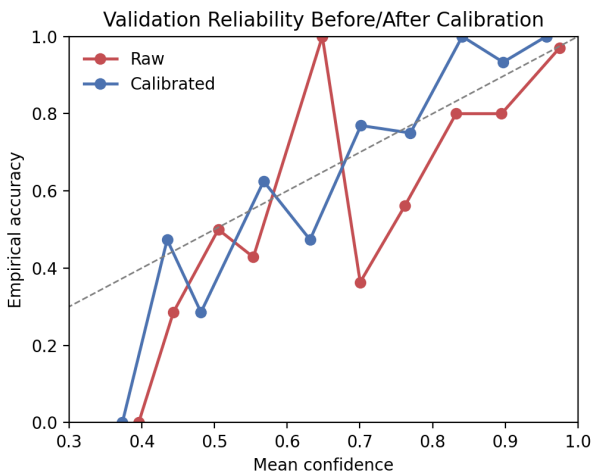

Figure 3. Validation reliability plot for the MLP before and after temperature scaling. The diagonal represents perfect calibration, where confidence matches observed correctness. The raw model deviates noticeably from this ideal line, indicating that some predictions are overconfident. After temperature scaling, the calibrated curve moves closer to the diagonal across much of the confidence range, showing that the reported probabilities better reflect true likelihood of correctness. This is consistent with the reduction in validation expected calibration error from 0.1042 to 0.0857.

selection.

Since temperature scaling is designed to improve the reliability of confidence estimates, not to optimize discrete class prediction directly, this small reduction in classification score does not necessarily imply a weakness in the model. The calibrated model may become slightly less sharp in its top-1 outputs while becoming more trustworthy as a basis for action thresholds. Since the use case is triage rather than autonomous legal determination, better confidence alignment is arguably more important than preserving every fraction of uncalibrated macro-F1.

Policy	Acc.	F1	AF Prec.	AF Rec.	FP AF
Threshold	0.4825	0.4809	–	–	–
Cal. MLP	0.6930	0.6926	–	–	–
0.80/0.55	–	–	0.882	0.375	2
0.70/0.70	–	–	0.800	0.400	4

Table 2. Policy trade-offs on the test set. AF = auto-flag.

To analyze policy behavior, the calibrated probabilities are converted into DMCA-style actions through confidence thresholds. Under a conservative policy with $\tau_{\text{auto}} = 0.80$ and $\tau_{\text{review}} = 0.55$, the model auto-flags 17 test cases, sends 68 to review, and assigns 29 to no action. At this operating point, auto-flag precision for true knockoffs is 0.882, auto-flag recall is 0.375, and there are only 2 false-positive auto-flags. A more permissive alternative with $\tau_{\text{auto}} = 0.70$ and $\tau_{\text{review}} = 0.70$ raises auto-flag recall to 0.400, but lowers auto-flag precision to 0.800 and increases false-positive auto-flags to 4.

These results suggest that the system is best interpreted as a calibrated triage mechanism rather than a hard classifier. The relevant trade-off is not only between classes, but between false-positive enforcement risk and automatic catch rate.

5.3. Discussion

Due to the nuances in ambiguous similarity versus knockoff cases in fashion infringement, creating a discrete 3-class classifier presents as a more intricate task than generic visual retrieval because the intermediate *similar* class cannot be captured reliably by a fixed threshold alone. The absolute-difference and nearest-original feature design appears to be effective, since the MLP substantially improves on both thresholding and logistic regression. Finally, cal-

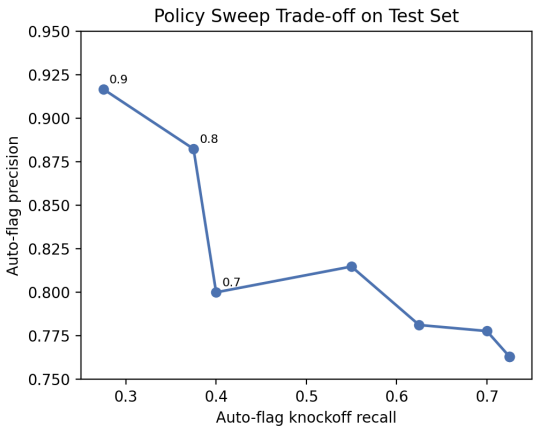

Figure 4. Policy-threshold trade-off for the calibrated decision layer. The horizontal axis shows auto-flag knockoff recall, that is, the proportion of true knockoff cases automatically caught by the system, while the vertical axis shows auto-flag precision, that is, the proportion of automatically flagged cases that are actually knockoffs. Each plotted point corresponds to a different confidence-threshold policy. Moving toward higher recall means the system catches more real knockoffs automatically, but usually at the cost of lower precision and therefore greater false-positive enforcement risk.

ibration matters for deployment framing. Even when the uncalibrated MLP has the best macro-F1, the calibrated model is more suitable for confidence-based review workflows.

Most difficult cases arise around the boundary between *similar* and *knockoff*, which is consistent with the semantics of the task rather than with a simple failure to capture gross visual similarity. Essentially, the classifier is being asked to resolve a category boundary that is itself subjective and partly legal in nature, so residual confusion in this region should be expected even when the broader three-way separation improves.

The confusion matrix helps make this concrete. For true *knockoff* cases, the calibrated model still confuses a minority of examples with both *similar* and *original*, suggesting that some copied designs do not appear uniformly close under every embedding-derived signal. For true *original* cases, the remaining false positives indicate that visual or semantic proximity to a nearest original can sometimes mimic infringement-like structure even when the label is ultimately non-infringing. The middle *similar* class remains the hardest overall because it is not a clear category. It contains borderline designs that partly resemble true knockoffs and partly resemble benign originals. This explains why the model benefits from richer pairwise features but still does not eliminate ambiguity.

From an error-analysis perspective, the failures are therefore meaningful rather than random. False-positive auto-flags correspond to the most operationally sensitive mistake, because they escalate non-knockoff cases unnecessarily. False negatives for the knockoff class represent the opposite risk: copied or highly derivative cases being routed only to review or to no action. The policy-threshold analysis shows that these error types can be traded off, but not removed entirely. This reinforces the interpretation of the system as a calibrated decision-support tool whose purpose is to structure review workload and risk exposure, not to produce definitive legal judgments. It is also important to distinguish the two evidence streams in the study. The triplet ResNet pipeline is the image-side representation-learning path, whereas the strongest reported Stage-2 quantitative results are currently obtained on stabilized CLIP text embeddings rather than on a fully end-to-end learned visual backbone.

There are also clear limitations. The current empirical evidence is strongest for the MLP decision layer, not for a full end-to-end comparison of learned embedding models. The labels themselves encode a legally and environmentally subjective boundary, especially between *similar* and *knockoff*, and the system should not be interpreted as making legal determinations. Finally, the dataset remains relatively small for a high-variance visual task, so broader generalization claims would require larger-scale and more systematically annotated evaluation data, ideally with stronger annotation protocols and external validation.

6. Related Work

Computer vision approaches to similarity matching have historically focused on retrieval, verification, and near-duplicate detection rather than on graded legal judgment. Earlier systems used hand-crafted local features and image matching pipelines to identify duplicated or highly overlapping images. These methods are effective when the task is exact or near-exact matching, but they are less suitable when similarity is partial, stylistic, or semantically ambiguous, as is often the case in fashion.

More recent work has shifted toward learned visual embeddings. Siamese and triplet architectures learn a representation space in which semantically similar inputs are close under a distance metric, enabling ranking and nearest-neighbor retrieval. Early Siamese-network work established shared-weight neural architectures for similarity learning (1), while FaceNet demonstrated the effectiveness of triplet loss for embedding learning in visual verification (2). Subsequent work argued that triplet-based metric learning can remain highly competitive when supported by appropriate mining strategies (10), and alternative loss formulations such as quadruplet loss were proposed to strengthen inter-class separation beyond standard triplet constraints (11). Later metric-learning work also emphasized that performance depends strongly on how informative positives and negatives are weighted or mined during training (16). Self-supervised contrastive methods such as SimCLR later showed that strong transferable image representations can be learned without dense manual labels (3). CLIP further demonstrated that large-scale contrastive pretraining can align visual and textual representations in a shared embed-

ding space (4).

In fashion vision specifically, datasets such as DeepFashion have enabled work on retrieval, attribute prediction, and consumer-to-shop matching (5). These studies show that neural representations can capture fine-grained cues such as texture, cut, silhouette, and detailing. However, the dominant evaluation settings in this literature are retrieval quality and binary matching rather than graded semantic or legal distinctions. As a result, existing fashion-vision work does not directly address the problem of distinguishing clearly distinct designs from stylistically similar ones and from likely knockoffs.

This study builds on that literature by treating similarity as an intermediate signal rather than the final output. The contribution lies in combining a deep representation-learning stage with a calibrated decision layer that turns pairwise similarity structure into three-way labels and review actions. This makes the work related not only to metric learning, but also to calibration and decision-support systems for high-ambiguity classification, including selective-prediction settings in which the model should act conservatively on low-confidence cases (7; 8).

7. Conclusions

This study shows that graded fashion infringement assessment is feasible as a three-class classification problem, but only when the similarity signal is coupled with a learned non-linear decision layer. Across the the experiments, fixed cosine thresholds performed poorly, while a pairwise MLP over embedding-derived features gave the strongest test performance. Calibration then turns that classifier into a more useful confidence-aware review tool (6).

The main lesson from the study is that separating representation learning from decision calibration is productive in this domain. Deep metric learning provides the conceptual and technical basis for learning fashion-aware embeddings (1; 2; 3), while the MLP decision layer makes the final task interpretable, analyzable, and tunable for risk-sensitive operation. The current strongest evidence therefore supports the calibrated decision-layer formulation, with metric-learning improvements remaining an important next step rather than an already-closed empirical question.

With respect to the research questions, the results support three main conclusions. First, embedding-based similarity does contain useful infringement signal, but not in a form that can be used directly through raw similarity alone: the representation supports the task only when paired with a learned decision layer that can distinguish between *original*, *similar*, and *knockoff* cases. Second, graded infringement decisions are highly sensitive to the form of the decision rule, since both fixed thresholds and linear baselines perform substantially worse than the non-linear MLP. Third, deep metric learning remains a justified representation-learning strategy for the project, but the strongest completed empirical evidence currently lies in the calibrated downstream classifier rather than in a full end-to-end comparison of learned visual backbones. The overall answer is therefore that a two-stage calibrated decision pipeline is effective for the present task, while improved representation learning remains the clearest direction for future work.

Future work should evaluate multiple learned embedding backbones end-to-end on the same downstream task, increase the scale and annotation consistency of the infringement dataset, and study whether richer multimodal signals such as textual descriptions or structured design attributes further improve separation between *similar* and *knockoff* cases.

Appendix

A. Additional Plots

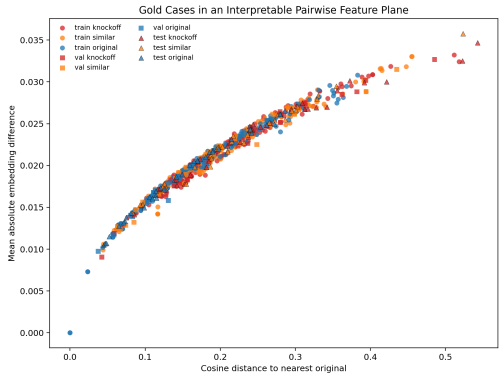

Figure 5. The visible class overlap shows that no single distance cutoff can cleanly separate the three categories: cases with similar pairwise distances can still belong to different labels, especially around the ambiguous *similar* region. This helps explain why fixed-threshold and linear baselines underperform relative to the non-linear MLP, which can model more complex interactions between pairwise features.

B. Additional Dataset and Label Notes

The downstream classifier used in the report contains 743 embedded examples split into 519 training, 110 validation, and 114 test cases. The class mapping is `knockoff` = 0, `similar` = 1, and `original` = 2. For training examples that are themselves labelled *original*, nearest-original feature construction excludes the item itself from the reference search, preventing trivial self-matches. This matters because otherwise some training instances would achieve artificially perfect nearest-neighbour comparisons.

An important limitation is that the representation-learning and decision-layer stages are not yet evaluated as a single unified end-to-end experimental stack on the same final infringement benchmark. The appendix therefore makes explicit that the report combines two kinds of evidence: image-based triplet-learning infrastructure for representation learning, and the strongest completed quantitative results from the Stage-2 calibrated classifier built on stabilized CLIP text embeddings.

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